UNIVERSITY OF TARTU
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Pangenome graph approaches for demographic inference from Estonian long-read genomes

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1. Pangenomics

The Human Pangenome Reference Consortium showed that human genetic diversity is incompletely represented by a single linear reference[1]. Its first draft pangenome, based on 94 haplotype-resolved assemblies from 47 ancestrally diverse individuals, added substantial polymorphic sequence relative to GRCh38 and improved discovery of both small and structural variants. This resource provides the conceptual basis for using graph-based references in human population-genetic analyses.

2. Research background

My research focuses on reconstructing human population history from genome-wide variation and uniparental genomes, specifically the Y chromosome and mitochondrial DNA[2-5]. These data provide complementary insights into demographic history, lineage continuity, population structure, and sex-specific processes.

3. Forthcoming long-read resource

The Institute of Genomics of the University of Tartu, will generate long-read PacBio whole-genome sequences from approximately 10,000 Estonian Biobank[6] gene donors[7]. This dataset provides an opportunity to study demographic history both within Estonia and in a broader European and global context, with improved access to complex genomic regions.

PacBio HiFi Revio™ sequencer in the Institute of Genomics of the University of Tartu

Image source: https://genomics.ut.ee/en/news/new-technology-help-sequence-whole-genomes-10000-gene-donors

4. Why linear references are limiting

Many loci that are informative for demographic inference can be difficult to recover when using a single linear reference genome[1,8]. This issue is particularly relevant for the Y chromosome and other repetitive or structurally complex regions[9-11].

5. Consequences for population genetics

Linear-reference workflows may lead to:

- reference bias;
- reduced detection of structural variants;
- incomplete reconstruction of extended haplotypes;
- loss of information from complex genomic regions.

Linear reference:

A — C — G — T — A — C — T

Pangenome graph:

A — C — G — T — A — C — T

\ G — A /

\ deletion /

\ insertion /

Graph-based references can reveal alternative haplotypes and structural variants that are difficult to capture with a single linear sequence.

6. Pangenome opportunity

Pangenome graphs extend the reference concept from a single linear sequence to a representation of multiple haplotypes and structural variants[12]. For population-genetic analyses, this matters because conclusions depend on which variants are detectable and how accurately haplotypes are reconstructed. Combined with long-read sequencing, graph-based references may improve access to lineage-informative variation that is underrepresented in linear-reference analyses.

7. Conceptual workflow

Estonian PacBio long-read genomes

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Variant discovery and haplotype reconstruction

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Graph-based representation of selected regions or representative genomes

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Graph visualisation and quality control

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Graph-based genotyping

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Demographic analysis using a broader spectrum of genome variation

8. Questions to discuss during the course

- Which pangenome graph strategies are realistic for thousands of long-read human genomes?
- How can graph-based genotyping be integrated with population-genetic analyses?
- Which classes of structural variation are most informative for population history?
- How can the Y chromosome and other complex regions be analysed with reduced reference bias?

Direct course relevance

The course programme is directly relevant because it covers the main methodological steps needed for this project: sequence alignment, genome assembly, pangenome graph representations, graph construction with PGGB, graph visualisation and manipulation with ODGI, implicit pangenome graphs, and genotyping from graph references with PanGenie and POVU.

Expected outcome

The expected outcome is a practical strategy for evaluating pangenome graph methods in demographic analyses of Estonian long-read genomes, including fine-scale variation within Estonia and comparisons with wider European and global datasets.

Main objective

Evaluate how pangenome graph methods can extend demographic inference beyond variants accessible through a single linear reference genome.

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